

Problem-Solution Fit Canvas 2.0

Purpose / Vision

Digitize metro ticket booking and validation through ServiceNow so passengers can book faster, receive accurate fares and obtain a verifiable QR-based digital ticket, while metro operations gain centralized, traceable and maintainable workflow automation.

1. CUSTOMER SEGMENT(S)

- Primary: Metro passengers / daily commuters using self-service digital booking.
- Secondary: Station managers and metro operations staff who validate and monitor tickets.
- Supporting user: ServiceNow administrators responsible for workflow, roles, ACLs and maintenance.

6. CUSTOMER CONSTRAINTS

- Passenger may have limited time during peak travel.
- Digital booking should remain simple for users with limited technical knowledge.
- Portal behaviour and QR rendering must work reliably within ServiceNow constraints.
- Fare logic and station mapping must be accurate to maintain trust.
- The first version does not yet include an integrated payment gateway, SMS/WhatsApp ticketing or analytics dashboard.

5. AVAILABLE SOLUTIONS

- Physical ticket counter: familiar but involves queues, manual handling and paper usage.
- Ticket vending / AFC systems: faster than counters but still requires station-side infrastructure.
- Existing mobile/web ticketing approaches: convenient but may exist outside the organization's ServiceNow workflow.
- Manual records or separate systems: workable for operations but create fragmented data and weaker traceability.

2. JOBS-TO-BE-DONE / PROBLEMS

- Book a metro ticket without depending completely on a physical counter.
- Select source, destination, passenger type and ticket quantity correctly.
- Know the correct fare before completing the booking.
- Receive reliable proof of booking that can be verified at the station.
- Track the submitted request and retain booking information digitally.
- For operations: reduce manual processing, data mismatch and repeated verification work.

9. PROBLEM ROOT CAUSE

- Traditional ticketing depends heavily on manual counters, paper and separate operational steps.
- Fare, booking data, workflow and validation are not always handled as one connected digital process.
- Manual intervention increases waiting time and creates opportunities for fare or data inconsistencies.
- Digital portal environments introduce additional integration challenges such as variable mapping and QR rendering compatibility.

7. BEHAVIOUR

- Passenger opens the ServiceNow portal and selects the metro booking catalog item.
- Enters journey details and reviews the calculated fare.
- Submits the request and checks status / ticket result.
- Opens the QR digital ticket and presents it to station staff.
- Operations users verify data, review booking records and use Flow Designer execution details when troubleshooting.

3. TRIGGERS

- Need to travel by metro immediately or during peak hours.
- Long queues or slow manual ticketing create a reason to choose self-service.
- Passenger wants a paperless ticket available on a phone.
- Station/operations teams need faster access to booking and validation data.

4. EMOTIONS: BEFORE / AFTER

- Before: rushed, uncertain about waiting time, concerned about fare accuracy, dependent on counter availability.
- During: expects a simple form, clear fare and visible confirmation.
- After successful booking: confident, in control, reassured by request status and QR proof.
- After successful validation: relieved that the ticket is accepted quickly without paper handling.

10. YOUR SOLUTION

- A ServiceNow-based Metro Ticket Generating System using Service Catalog for passenger booking.
- Dynamic fare processing based on source, destination and ticket details.
- Flow Designer for end-to-end automation and record creation.
- Custom Metro Database table for structured booking storage.
- QR-based digital ticket rendered through a Service Portal-compatible widget / spModal approach.
- Request tracking, ACL / role-based access and Flow Execution Details for operations and troubleshooting.
- Future scope: payment gateway, email/SMS/WhatsApp delivery and travel analytics.

8. CHANNELS OF BEHAVIOUR

- Online: ServiceNow Service Portal, catalog item, request status, QR widget / spModal, future email/SMS channels.
- Offline: metro station, ticket-validation point, interaction with station staff, physical travel environment.

PROBLEM-SOLUTION FIT SUMMARY

- Problem fit: The solution directly targets queue dependency, manual handling, fare/data inconsistency and weak digital ticket validation.
- Behaviour fit: The passenger journey already follows a natural digital sequence - enter journey details, review fare, submit, receive ticket, present for validation.
- Constraint fit: The design stays inside ServiceNow using Service Catalog, Flow Designer, custom table, UI policies/client scripts, ACLs and a portal-compatible QR presentation.
- Current scope discipline: payment gateway, SMS/email/WhatsApp delivery and analytics are treated as future enhancements rather than implemented features.